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•repulsive force balanced by components of attractive forces between electrons and nuclei.

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3. Without invoking time, what can we find out about changes in molecular structure from simple approximations?

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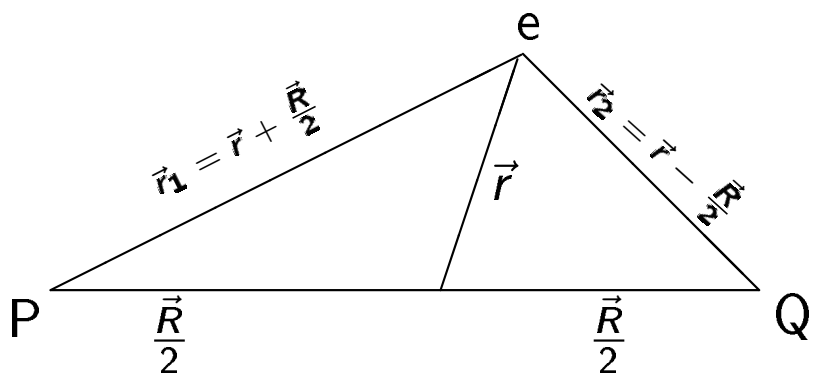
Find electronic energy levels $E_n(\vec{R})$.

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$$\text{as } R \longrightarrow 0 \quad \mathcal{E} = \frac{e^2}{R} - 2^2 \frac{e^2}{2a_0}$$

*wavefunction goes to hydrogenic* $Z = 2$ *limit*.

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A good trial function: *LCAO-MO approximation*

$$v \longrightarrow \frac{1}{\sqrt{2}}[u(\vec{r}_1) + u(\vec{r}_2)]$$

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*This is called the LCAO-MO approximation (LCAO=Linear combination of atomic orbitals)*

*consider complete Hamiltonian*

$$\hat{H} = \hat{T}_N + \hat{T}_e + V(\vec{r}, \vec{R})$$

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*R is now a parameter*
*and* $\mathcal{E}_n(\vec{R})$ *is then a potential energy surface (PES) for nuclear motion*
*: also called adiabatic PES*

*Consider basis set expansion of* $\Psi(\vec{r},\vec{R})$ *in terms of* $v_n(\vec{r},\vec{R})$ *functions,*

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***Born-Oppenheimer Approximation***

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*Above equation is the eigenvalue problem for the nuclear motion.*

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$$\phi(\vec{R}) = \frac{1}{R}f(R)Y_j^m(\Omega); \quad \nabla_R^2 = \frac{1}{R}\frac{\partial^2}{\partial R^2}R - \frac{\hat{J}^2}{\hbar^2 R^2}$$

*For given j :* $\hat{J}^2\phi(\vec{R}) = \hbar^2 j(j+1)\phi(\vec{R})$

$$-\frac{\hbar^2}{2M}\frac{d^2fY}{dR^2} + \left\{\mathcal{E}_0 + \frac{1}{2}M\omega^2(R-R_0)^2 + \frac{\hbar^2 j(j+1)}{2MR^2}\right\}fY = EfY$$

or, dividing both sides by Y,

*consider a diatomic molecule again.*

*Suppose we solve the elctronic problem. In particular, consider ground state energies,* $\mathcal{E}_{grd.}(\vec{R}) \equiv \mathcal{E}(\vec{R})$

*Turn nuclei loose in this 'potential'.*

*eigen value problem for nuclei:* $\left[-\frac{\hbar^2}{2M}\nabla_R^2 + \mathcal{E}(\vec{R})\right]\phi(\vec{R}) = E\phi(\vec{R})$

$$\text{expand around minimum} : \mathcal{E}(\vec{R}) \approx \mathcal{E}_0 + \frac{1}{2}\left.\frac{\partial^2\mathcal{E}}{\partial R^2}\right|_{R_0}(R-R_0)^2 + .....$$

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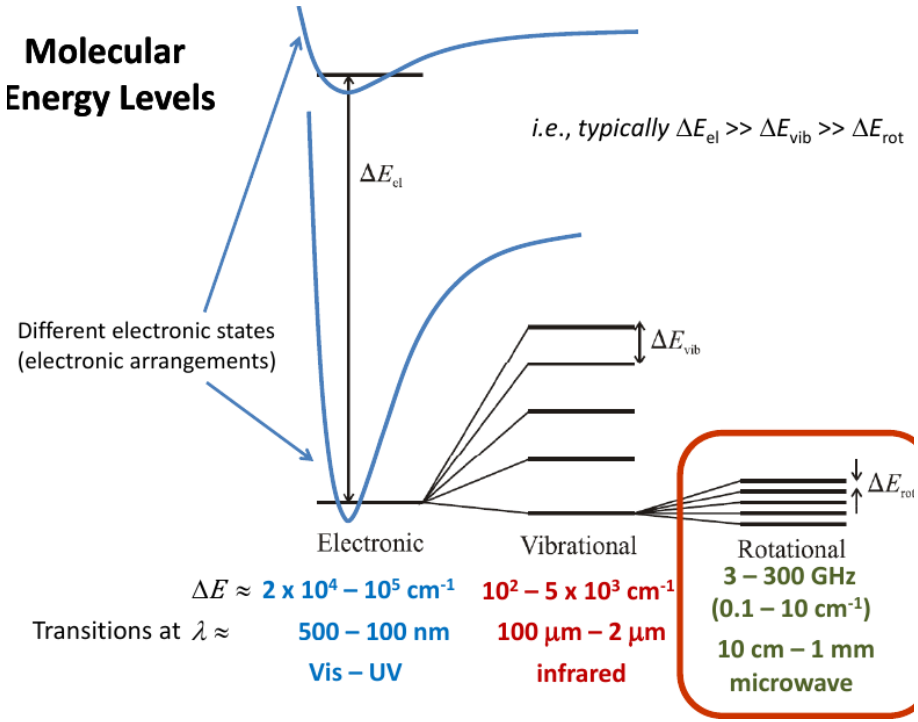
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*1st term : electronic, 2nd : vibrational,*

*and 3rd : rotational part of total energy*



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*where the exact result is:*

$\mathcal{E}_0 = -\frac{e^2}{2a_0} - 2.8eV$ *at* 106.3 *pm, where* $-\frac{e^2}{2a_0}$ *is the energy of the free H atom*